

Problem Sheet 7

1.

(a) State the integral form of the Laplace transform.

(b) Apply it to evaluate the transform of the following functions:

(i) $f(t) = 8u(t)$

(ii) $f(t) = 4e^{-at}$

(iii) $f(t) = u(t) + e^{-t}u(t)$

2.

(a) Prove the linearity property of the Laplace Transform.

(b) Use the Laplace Transform to find a solution for the differential equation below, assuming that the system is initially relaxed:

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = \delta(t).$$

(c) Solve the equation:

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 0,$$

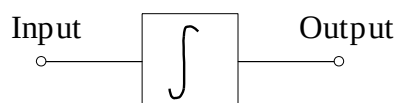
with initial conditions

$$y(0^+) = 5$$

$$\frac{dy(0^+)}{dt} = 0.$$

3. SEMINAR QUESTION I

(a) Define the transfer function in terms of Laplace transforms of the input and output of a system. What is the transfer function of a perfect voltage amplifier with a gain of 40 dB? What is the impulse response function in the time domain?

(b) How do we use the transfer function in instrumentation analysis to find the output of a system in time in response to an arbitrary forcing function $y(t)$? What properties of the delta function or impulse $\delta(t)$ make it particularly important for this application?(c) An integrator unit is fed a $\delta(t)$ input at a time $t = 0$. What is the output of the unit?

(d) Write down the Laplace transforms for the input and output of this device and use your results to find the transfer function of the integrator.

4. SEMINAR QUESTION II

- (a) Two identical amplifiers with impulse response $g(t) = te^{-t}$ in the time domain are connected in series. What is the transfer function of a single amplifier in isolation?
- (b) What is the transfer function of the two amplifiers in series?
- (c) Using partial fractions find the time domain output of the two amplifiers in series for a step function input $y(t) = u(t)$ at a time $t = 0$.

Hints and Numerical Answers

1 (b) (iii) $\frac{2s+1}{s(s+1)}$

2 (a) Use the fact that integration is a linear operation (b) $e^{-t} - e^{-2t}$ (c) $10e^{-t} - 5e^{-2t}$

3 (a) 100, $100\delta(t)$ (c) 1 (d) $1/s$

4 (c) $u(t) - e^{-t} - te^{-t} - \frac{t^2 e^{-t}}{2!} - \frac{t^3 e^{-t}}{3!}$